

ENTERPRISE E-COMMERCE INTELLIGENCE & CUSTOMER ANALYTICS PLATFORM

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Core Technologies Used: Python (Pandas, NumPy), Advanced Excel, MySQL, Power BI

1. Executive Summary :

This report delivers a comprehensive data analysis of the Brazilian Olist e-commerce public dataset, covering transactions from 2016 through 2018. Utilizing an end-to-end data analytics workflow—spanning data acquisition, cleaning, SQL querying, statistical evaluation, and interactive dashboarding—this project examines overall sales performance, customer purchasing behaviour, product category demand and seller contributions. The objective is to translate complex transactional data into clear, actionable business insights suitable for executive decision-making and portfolio presentation.

2. Key Performance Metrics Summary:

- **Total Revenue : R\$ 13,591,644.00**
- **Total Unique Orders: 98,666**
- **Total Unique Customers (customer_unique_id): 96,096**
- **Average Order Value (AOV): R\$ 137.75**
- **Total Order Items: 112,650**
- **Total Freight Value: R\$ 2,251,909.54**
- **Repeat Customers: 2,997**
- **One-Time Customers: 93,099**
- **Repeat Customer Rate: 3.12%**
- **Total Delivered Orders: 96,476**
- **On-Time Delivered Orders: 88,649**
- **Late Delivered Orders: 7,827**
- **On-Time Delivery Rate: 91.89%**
- **Average Delivery Time: 12.56 days**

3. Core Business Findings & Analysis:

3.1 Sales Performance:

The platform generated R\$ 13.59 million in item-price revenue across 98,666 unique orders. The average order value was R\$ 137.75, with an average of 1.14 items per order. These metrics provide a baseline for evaluating sales performance and exploring cross-selling opportunities.

3.2 Customer Behaviour:

The analysis identified 96,096 unique customers, of whom 2,997 were repeat customers, resulting in a repeat customer rate of 3.12%. This provides a baseline for further customer retention analysis.

3.3 Product Performance:

Product and category analysis was conducted to examine revenue contribution and sales volume. These findings can support product assortment, demand analysis and inventory planning.

3.4 Delivery Performance:

Of the 96,476 delivered orders, 91.89% were delivered on time. The average delivery time was 12.56 days. These metrics provide a basis for monitoring logistics performance and investigating delivery delays.

3.5 Regional and Seller Analysis:

State-wise customer distribution, order activity, revenue contribution and seller performance were examined using SQL and dashboard analysis. These insights can support regional performance monitoring and seller contribution analysis.

3.6 Statistical Analysis:

Statistical analysis showed a mean item price of R\$ 120.65, a median of R\$ 74.99 and a mode of R\$ 59.90. The price-freight correlation was 0.4142, indicating a moderate positive association. This does not establish causation.

4. Strategic Business Recommendations:

- **Bring Back First-Time Buyers:** Only about 3% of customers buy more than once. We should use automatic reminder emails, special product suggestions, and discount coupons to encourage single-time buyers to shop again.
- **Fix Late Deliveries:** Even though most orders arrive on time (about 92%), we still had over 7,800 late shipments. We need to look closely at our shipping routes and delivery partners to figure out what went wrong and fix those delays.
- **Focus on Our Best-Selling Products:** We should put more money and advertisements into our top-performing product categories to maximize our sales and returns.

5. Conclusion:

This project demonstrates an end-to-end data analytics workflow using Python, Excel, MySQL, statistical analysis and Power BI. It provides insights into sales, customer behaviour, product demand, regional activity, seller contributions and delivery performance. The findings establish measurable business baselines and demonstrate how data analytics can support informed business decision-making.